

ATUL KUMAR

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EDUCATION

Birla Institute of Technology, Mesra <i>Bachelor of Technology in Chemical Engineering</i>	2023 – Present Ranchi, India
Chinmaya Vidyalaya, Bokaro Steel City <i>Higher Secondary: Class: 11,12</i>	2021 – 2022 Bokaro, Jharkhand

EXPERIENCE

Web Development Intern <i>Zaalima Development (Remote)</i>	Apr 2026 – Jun 2026
- Developed an AI-powered Chrome Extension Generator using React, TypeScript, and Tailwind CSS, automating the generation of 5+ core extension files from natural language prompts and reducing project setup time by 80%. - Implemented a modular code generation engine supporting 20+ Chrome Extension APIs with reusable templates and validation, enabling rapid creation of production-ready browser extensions. - Built a RAG-based enterprise knowledge assistant using FastAPI and LangChain with a 5-stage retrieval pipeline for intelligent SOP search and question answering. - Implemented semantic search and vector embeddings to deliver 100% source-grounded responses, improving retrieval accuracy and reducing LLM hallucinations.	

PROJECTS

Stock-Research-Assistant <i>Python, FastAPI, LLMs, Financial APIs</i> 🔄	Jul 2026
- Developed an AI-powered Stock Research Assistant integrating 4+ analysis modules for technical, fundamental, and sentiment analysis. - Built a FastAPI backend with 10+ REST API endpoints for automated stock data processing and AI-driven research reports. - Implemented LLM-based summarization analyzing 8+ financial indicators, reducing manual research time by 70%. - Designed a modular architecture integrating external APIs for real-time market data and explainable insights.	
SOP-RAG-Assistant <i>FastAPI, MongoDB, LLMs, Vector Embeddings</i> 🔄	May 2026
- Built an enterprise RAG assistant using FastAPI and LangChain with a 5-stage retrieval pipeline for SOP search. - Implemented semantic search using vector embeddings, improving response relevance by 40% over keyword-based search. - Developed an end-to-end document processing pipeline including PDF parsing, chunking, and embedding generation, reducing retrieval latency by 35%. - Scalable REST APIs with source-grounded responses, achieving 100% citation-backed AI-generated answers.	
Dental Caries Segmentation <i>YOLOv8, PyTorch, Computer Vision</i> 🔄	Jan 2026
- Developed a YOLOv8-based segmentation pipeline for detecting dental caries from 750+ radiographic images. - Trained on 768×768 images using AdamW optimizer with cosine learning-rate scheduling. - Applied augmentation techniques to increase effective dataset diversity by approximately 4×. - Achieved 22.6% mask recall using Precision, Recall, and mAP evaluation metrics.	

TECHNICAL SKILLS

Programming: Python, C, C++
Machine Learning: Supervised & Unsupervised Learning, Regression, Classification, Clustering, Model Evaluation
Libraries: Scikit-learn, NumPy, Pandas, PyTorch
Databases & Tools: MySQL, DBMS, Power BI, MS Excel, Tableau
Technologies: Git, Docker, Jupyter Notebook
Web Development: HTML, CSS, JavaScript, WordPress
Cloud: AWS, Google Cloud

CERTIFICATIONS & JOB SIMULATIONS

Deloitte Data Analytics Job Simulation (Forage)
Data Analyst Bootcamp (Udemy)